The Trees of New England: How Extreme Climate Change Will Impact the Region

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The Problem

"New England is now heading towards being like the south-eastern US." - Stephen Young

The New England region is experiencing some of the strongest climate change in the world. Rising temperatures can:

- Decrease soil moisture
- Bolster insect populations
- Shorten growing seasons

All of which endanger tree populations.

Fagus grandifolia (American Beech). Source: University of Toronto Scarborough

The importance of trees

Trees are often foundation species in forests.

- Trees absorb CO_2 from the atmosphere, interact with coastal and inland ecosystems, and serve as economic drivers through logging.

For example, Eastern Hemlock is a foundation species for forests in the Northeastern U.S. (like Harvard Forest).

- Research shows that the disappearance of Eastern hemlock could be catastrophic for stream ecosystems in the Northeast.

Eastern hemlock branches. Photo by Sara Lissaker via iStock

Data

Melillo et al. 2023

“Effects of Warming on Tree Species Recruitment at Harvard Forest and Duke Forest since 2009”

→ Controlled experiment altering air temperature within closed “chambers” in Harvard Forest

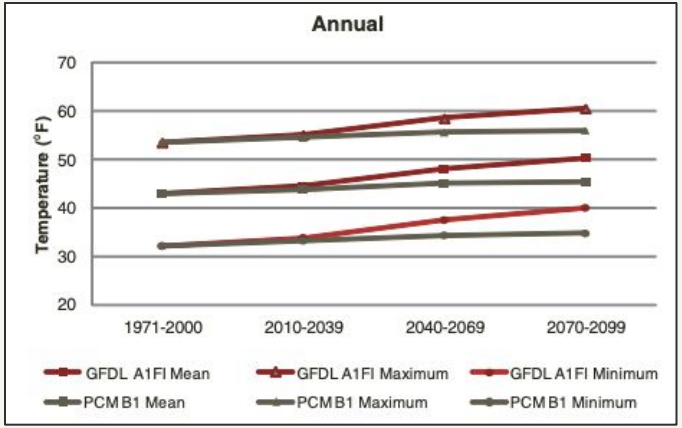
Record and Ellison 2023

“Bayesian Analysis of Tree Distributions Across Space and Time in Eastern North America 2010-2011”

→ Survey of tree counts (3 species) in a wide grid across the eastern U.S. and Canada.

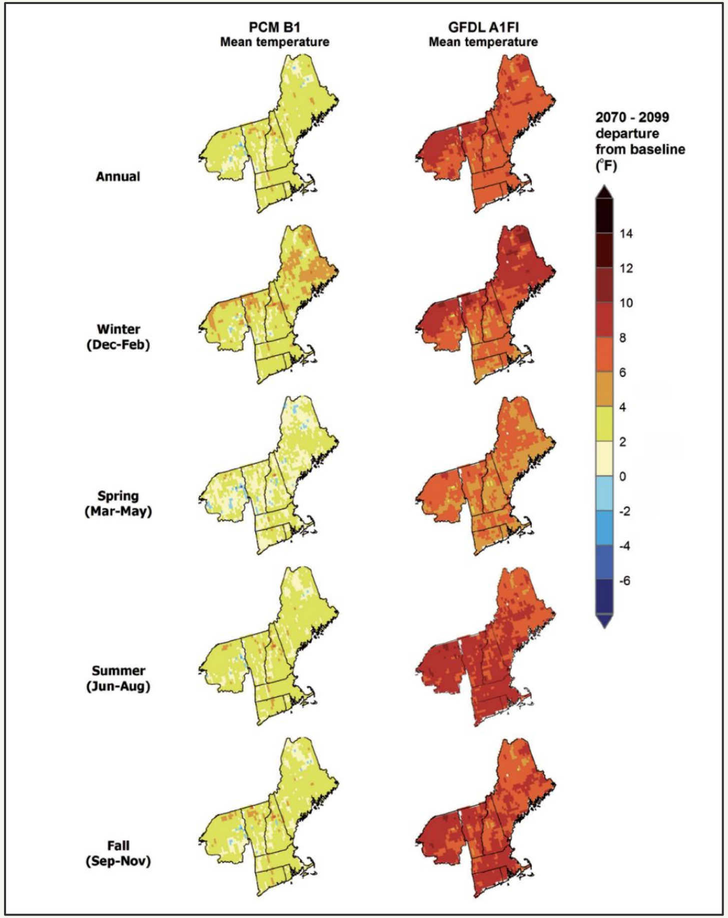
Projected Climate Outcome

PCM B1 and GFDL A1FI are two potential climate outcomes for the Northeast, with the former being a "low emission scenario" and the latter being a "high emission scenario."



(Janowiak et al, 2018)

Projected mean, minimum, and maximum temperature (°F) in the assessment area averaged over 30-year periods for the entire year and by season for two climate model-emissions scenario combinations. The 1971 to 2000 value is based on observed data from weather stations.



(Janowiak et al, 2018)

Projected difference in mean daily mean temperature (°F) at the end of the century (2070 through 2099) compared to baseline (1971 through 2000) for two climate model-emissions scenario combinations.

Results

Data: Record and Ellison 2023

Now

Hemlock and Beech trees prefer temperate climates with consistent, year-round precipitation.

Trees are not found in higher temperature areas.

Hemlocks are more widespread and exist in areas where precipitation can be lower than usual and/or less consistent, but are still very dependent on temperature.

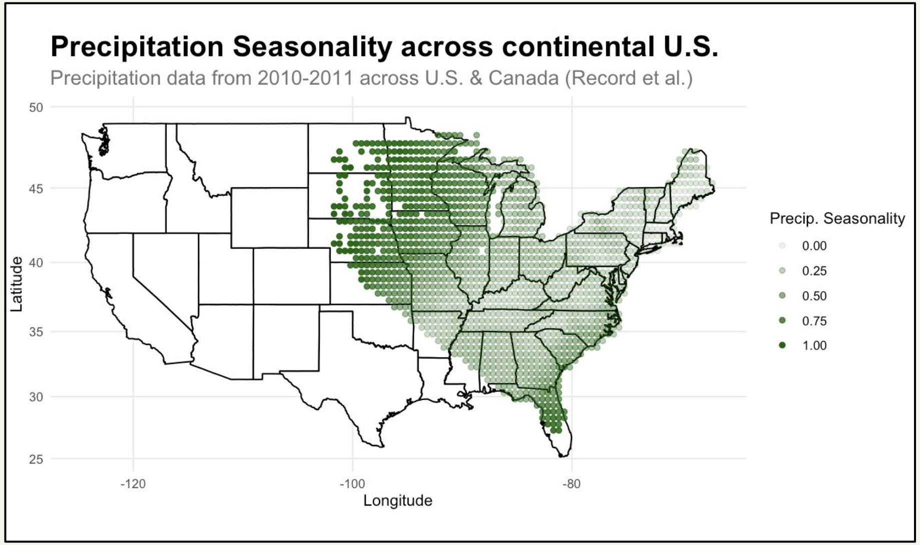
Moving forward

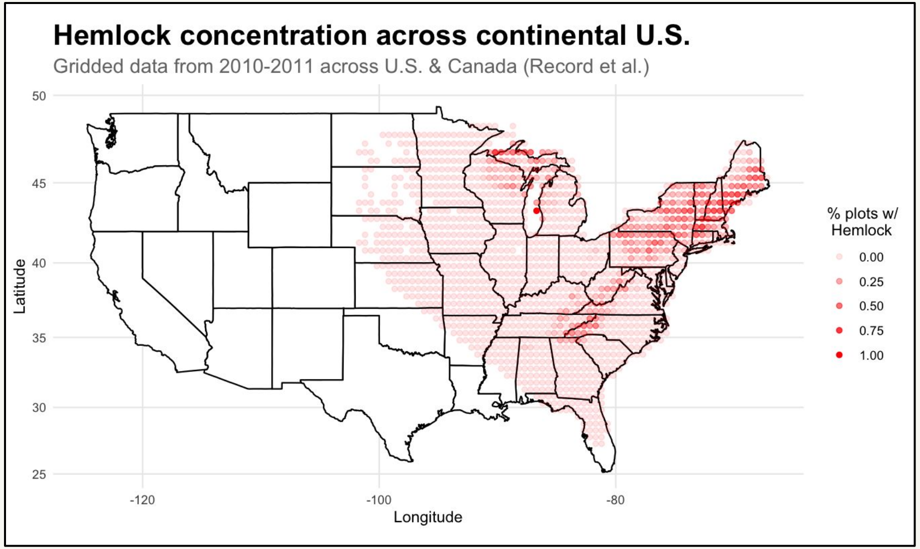
Temperature is a major factor for both of these species. They will be increasingly at risk as forest areas become hotter.

As precipitation and temperatures fluctuate more with climate change (Janowiak et al, 2018), these species could be seriously affected.

Impact of Precipitation Seasonality on Tree Growth

Eastern Hemlock is drought intolerant and was ranked as one of the 5 least adaptive species for climate change (Janowiak et al, 2018).





Left: Precipitation Seasonality Index (scaled using min-max method to achieve 0-1 range) plotted across geographical plot data. Right: Hemlock presence in plots divided by total number of plots in gridded region.

Surprising Findings

Data: Melillo et al. 2023

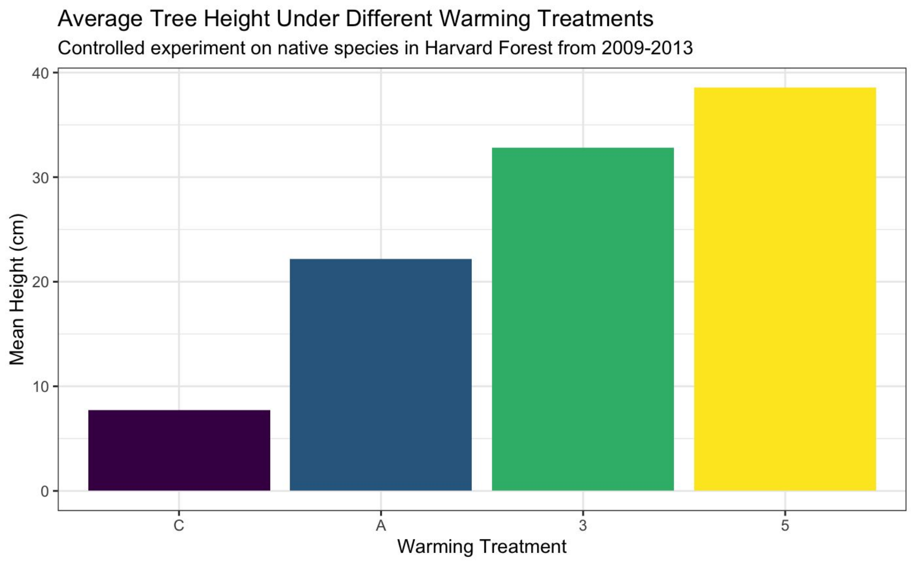
Height and diameter have a positive correlation with warming treatment

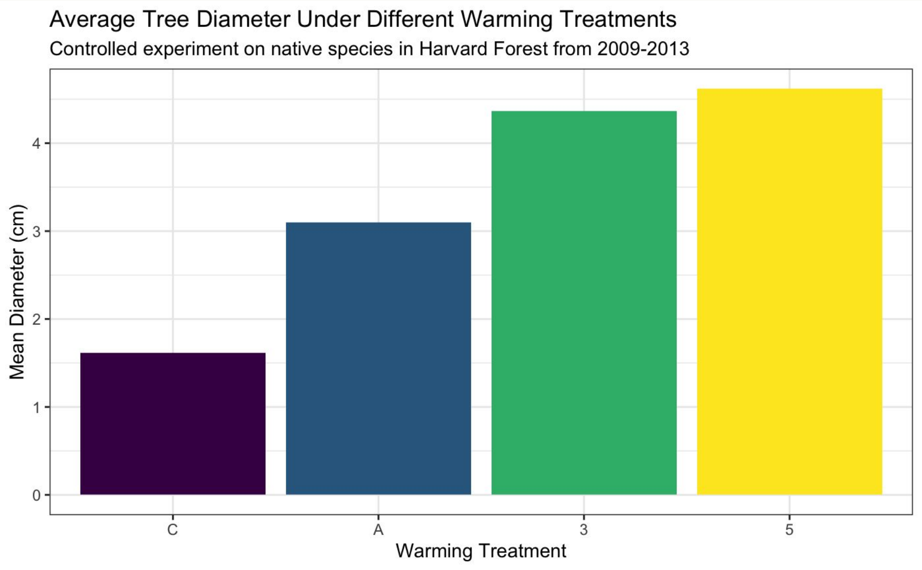
Lifespan and water stress have no correlation with warming treatment

These findings don’t support the Record et al. predictions!

Data: Melillo et al. 2023

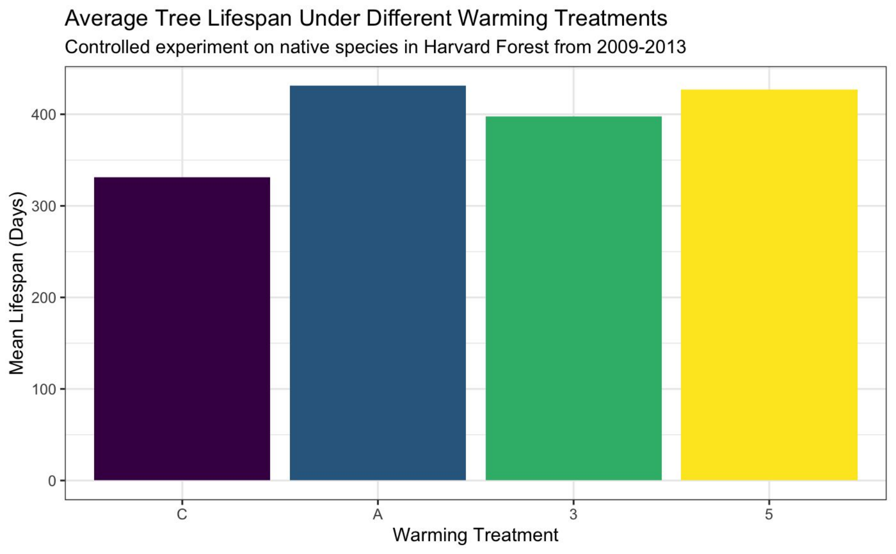
Height and diameter have a positive correlation on average with warming treatment

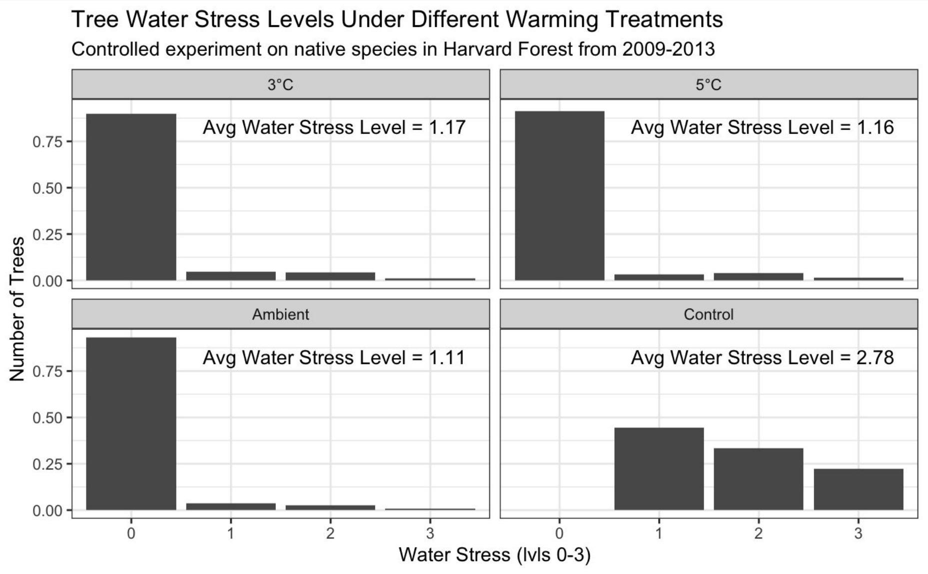




Data: Melillo et al. 2023

Lifespan and water stress have no correlation with warming treatment





Experiment vs. Survey

Data: Melillo et al. 2023

- Closed chambers don't have the same knock-on effects of temperature increase like increase in bug populations - no temperature increase in the wider ecosystem

- Unknown how precipitation was handled in closed chambers

Overall: Experiments done in a vacuum ≠ actual ecosystems!

Overall Conclusions

- Beech and Hemlock trees are both susceptible to climate change effects
- In a vacuum, heat and water effects do not have a major impact on tree growth
- Consistency of climatic variables is incredibly important — increasing fluctuations pose danger

Next Steps

- Former keystone species are declining and may not be resilient to future heat and precipitation fluctuations.
- More research needs to be done to identify alternate species that will be more resilient.

- Further inquiry can be done to identify the specific knock-on effects of climate change beyond basic climatic variables.
- Examples include the prevalence of bugs like the Hemlock Woolly Adelgid.

- Experiments done in the lab are only so helpful — tests on species resilience and introduction of new species should be done and monitored in the wild.

Thank You